

PERFORMANCE COMPARISON REPORT ON ADVANCED MACHINE LEARNING OPTIMIZATION

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Date: 6th June, 2026

1.0 Problem Statement

House prices are influenced by several factors such as property size, number of rooms, location and available facilities. While Linear Regression can be used to predict house prices, more advanced machine learning models may improve prediction accuracy.

The aim of this project was to compare the performance of different machine learning models for house price prediction. The models tested were Linear Regression, Decision Tree, Random Forest, and Gradient Boosting. Hyperparameter tuning using GridSearchCV was also applied to determine whether model performance could be improved.

2.0 Dataset Description

The Housing Prices Dataset was used for this analysis. The target variable was price, which represents the market value of a house. The dataset contained several housing features that may influence price, including area, bedrooms, bathrooms, stories, main road access, air conditioning, parking availability and preferred area.

These features were used to train and evaluate different machine learning models for predicting house prices.

1. **Area:** The total internal and external space footprint measured in square feet.
2. **Bedrooms:** The total number of designated sleeping quarters within the property.
3. **Bathrooms:** The total count of sanitary and bathing facilities.
4. **Stories:** The structural vertical height of the residential property.
5. **Main Road Access (mainroad):** A categorical indicator specifying proximity to primary transportation arteries.
6. **Air Conditioning (airconditioning):** A categorical indicator verifying the presence of centralized cooling climate systems.
7. **Parking Availability (parking):** Numeric capacity mapping vehicle storage capability.
8. **Preferred Area (prefarea):** A localized parameter identifying premium geographic positioning within the urban grid.

2.1 Data Preprocessing

The dataset was checked and prepared for machine learning analysis. Categorical variables such as "yes" and "no" were converted into numerical values using one-hot encoding. Feature scaling was applied using StandardScaler and the data was divided into training data (80%) and testing data (20%).

3.0 Models Implemented

Five independent model training and optimization phases were executed to map the predictive limits of the dataset:

1. **Baseline Linear Regression:** A traditional parametric estimator designed to map a uniform global hyperplane through the dataset by minimizing ordinary least squares.
2. **Decision Tree Regressor:** A non-parametric model that builds sequential conditional splits down branches to isolate distinct local neighborhoods of housing values.
3. **Random Forest Regressor:** An ensemble bagging architecture that trains hundreds of deep decision trees independently on random bootstrapped data subsets and averages their predictions to neutralize individual tree errors.
4. **Gradient Boosting Regressor:** An ensemble boosting framework that fits tree models sequentially, where each new tree is explicitly engineered to minimize the residual mathematical errors left behind by previous learners.
5. **Tuned Gradient Boosting (GridSearchCV):** An optimized boosting model where hyperparameters were systematically balanced using a 5-fold cross-validation grid search to isolate the ideal bias-variance equilibrium.

4.0 Evaluation Metrics

Because housing valuation is a continuous regression task, performance was evaluated by measuring the absolute mathematical distance between the model's predicted price and the true asset price across four industry-standard evaluation parameters:

1. **Mean Absolute Error (MAE):** Calculates the average absolute magnitude of the prediction errors, representing the standard expected price drift in original currency units without penalizing specific error scales.
2. **Mean Squared Error (MSE):** Averages the squared differences between predicted and actual values. This heavily penalizes massive prediction errors, making it highly sensitive to extreme data deviations.
3. **Root Mean Squared Error (RMSE):** The square root of the MSE, translating the exponential variance penalty scale back into the original currency units of the housing market for clear, intuitive interpretation.
4. **Coefficient of Determination (R2 Score):** Measures the proportion of total variance in housing prices that is successfully explained and predicted by the independent features in the model, scaling on a 0 to 1 spectrum.

5.0 Performance Comparison Matrix

The precise empirical outputs generated during the testing phase across all five modeling iterations are compiled in the structured matrix below:

S/N	Model Iteration	Mean Absolute Error (MAE)	Root Mean Squared Error (RMSE)	Coefficient of Determination (R ² Score)
1.	Baseline (Linear Reg)	1,029,010.29	1,381,117.65	0.622622
2.	Decision Tree	1,358,952.60	1,842,739.80	0.328194
3.	Random Forest	1,065,957.23	1,443,776.77	0.587603
4.	Gradient Boosting	1,040,051.69	1,439,009.70	0.590321
5.	Tuned Gradient Boosting	1,064,053.94	1,433,722.45	0.593326

6.0 Key Findings

1. The Linear Regression model produced the best overall results. It achieved the lowest RMSE (1,381,117.65) and the highest R² Score (0.6226), indicating that it explained about 62% of the variation in house prices.
2. The Decision Tree model had the weakest performance, with the highest prediction error and the lowest R² Score. This suggests that the model may have overfitted the training data.
3. Both Random Forest and Gradient Boosting performed better than the Decision Tree model. Hyperparameter tuning further improved the Gradient Boosting model, reducing prediction error slightly and increasing the R² Score.
4. The results suggest that the relationship between the selected housing features and house prices is largely linear, which explains why Linear Regression performed best.

7.0 Conclusion

This project compared several machine learning models for house price prediction. Although advanced models such as Random Forest and Gradient Boosting performed reasonably well, Linear Regression achieved the best overall performance on this dataset.

Recommendations

1. Linear Regression is recommended for house price prediction using the current dataset because it provides the highest accuracy and is easy to interpret unseen testing samples.
2. For immediate, resource-efficient forecasting within this current data scope, the Baseline Linear Regression model should be deployed as the primary predictive engine.
3. For future work, larger datasets and additional housing features could be included to determine whether more advanced machine learning models can outperform Linear Regression.